

Calculator Datasheet

Voltage (V)	Current & Resistance	$V = I * R$
	Power & Current	$V = P / I$
	Power & Resistance	$V = \text{Square Root of } (P * R)$
Current (I)	Voltage & Resistance	$I = V / R$
	Power & Voltage	$I = P / V$
	Power & Resistance	$I = \text{Square Root of } (P / R)$
Resistance (R)	Voltage & Current	$R = V / I$
	Voltage & Power	$R = (V * V) / P$
	Power & Current	$R = P / (I * I)$
Power (P / Watts)	Voltage & Current	$P = V * I$
	Current & Resistance	$P = I * I * R$
	Voltage & Resistance	$P = (V * V) / R$

Converter Datasheet

	Exa	Peta	Tera	Giga	Mega	Kilo	Base	Milli	Micro	Nano	Pico	Femto	Atto
Exa	1	$10^{\{3\}}$	$10^{\{6\}}$	$10^{\{9\}}$	$10^{\{12\}}$	$10^{\{15\}}$	$10^{\{18\}}$	$10^{\{21\}}$	$10^{\{24\}}$	$10^{\{27\}}$	$10^{\{30\}}$	$10^{\{33\}}$	$10^{\{36\}}$
Peta	$10^{\{-3\}}$	1	$10^{\{3\}}$	$10^{\{6\}}$	$10^{\{9\}}$	$10^{\{12\}}$	$10^{\{15\}}$	$10^{\{18\}}$	$10^{\{21\}}$	$10^{\{24\}}$	$10^{\{27\}}$	$10^{\{30\}}$	$10^{\{33\}}$
Tera	$10^{\{-6\}}$	$10^{\{-3\}}$	1	$10^{\{3\}}$	$10^{\{6\}}$	$10^{\{9\}}$	$10^{\{12\}}$	$10^{\{15\}}$	$10^{\{18\}}$	$10^{\{21\}}$	$10^{\{24\}}$	$10^{\{27\}}$	$10^{\{30\}}$
Giga	$10^{\{-9\}}$	$10^{\{-6\}}$	$10^{\{-3\}}$	1	$10^{\{3\}}$	$10^{\{6\}}$	$10^{\{9\}}$	$10^{\{12\}}$	$10^{\{15\}}$	$10^{\{18\}}$	$10^{\{21\}}$	$10^{\{24\}}$	$10^{\{27\}}$
Mega	$10^{\{-12\}}$	$10^{\{-9\}}$	$10^{\{-6\}}$	$10^{\{-3\}}$	1	$10^{\{3\}}$	$10^{\{6\}}$	$10^{\{9\}}$	$10^{\{12\}}$	$10^{\{15\}}$	$10^{\{18\}}$	$10^{\{21\}}$	$10^{\{24\}}$
Kilo	$10^{\{-15\}}$	$10^{\{-12\}}$	$10^{\{-9\}}$	$10^{\{-6\}}$	$10^{\{-3\}}$	1	$10^{\{3\}}$	$10^{\{6\}}$	$10^{\{9\}}$	$10^{\{12\}}$	$10^{\{15\}}$	$10^{\{18\}}$	$10^{\{21\}}$
Base	$10^{\{-18\}}$	$10^{\{-15\}}$	$10^{\{-12\}}$	$10^{\{-9\}}$	$10^{\{-6\}}$	$10^{\{-3\}}$	1	$10^{\{3\}}$	$10^{\{6\}}$	$10^{\{9\}}$	$10^{\{12\}}$	$10^{\{15\}}$	$10^{\{18\}}$
Milli	$10^{\{-21\}}$	$10^{\{-18\}}$	$10^{\{-15\}}$	$10^{\{-12\}}$	$10^{\{-9\}}$	$10^{\{-6\}}$	$10^{\{-3\}}$	1	$10^{\{3\}}$	$10^{\{6\}}$	$10^{\{9\}}$	$10^{\{12\}}$	$10^{\{15\}}$
Micro	$10^{\{-24\}}$	$10^{\{-21\}}$	$10^{\{-18\}}$	$10^{\{-15\}}$	$10^{\{-12\}}$	$10^{\{-9\}}$	$10^{\{-6\}}$	$10^{\{-3\}}$	1	$10^{\{3\}}$	$10^{\{6\}}$	$10^{\{9\}}$	$10^{\{12\}}$
Nano	$10^{\{-27\}}$	$10^{\{-24\}}$	$10^{\{-21\}}$	$10^{\{-18\}}$	$10^{\{-15\}}$	$10^{\{-12\}}$	$10^{\{-9\}}$	$10^{\{-6\}}$	$10^{\{-3\}}$	1	$10^{\{3\}}$	$10^{\{6\}}$	$10^{\{9\}}$
Pico	$10^{\{-30\}}$ ↓	$10^{\{-27\}}$	$10^{\{-24\}}$	$10^{\{-21\}}$	$10^{\{-18\}}$	$10^{\{-15\}}$	$10^{\{-12\}}$	$10^{\{-9\}}$	$10^{\{-6\}}$	$10^{\{-3\}}$	1	$10^{\{3\}}$	$10^{\{6\}}$
Femto	$10^{\{-33\}}$	$10^{\{-30\}}$ ↓	$10^{\{-27\}}$	$10^{\{-24\}}$	$10^{\{-21\}}$	$10^{\{-18\}}$	$10^{\{-15\}}$	$10^{\{-12\}}$	$10^{\{-9\}}$	$10^{\{-6\}}$	$10^{\{-3\}}$	1	$10^{\{3\}}$
Atto	$10^{\{-36\}}$ ↓	$10^{\{-33\}}$	$10^{\{-30\}}$ ↓	$10^{\{-27\}}$	$10^{\{-24\}}$	$10^{\{-21\}}$	$10^{\{-18\}}$	$10^{\{-15\}}$	$10^{\{-12\}}$	$10^{\{-9\}}$	$10^{\{-6\}}$	$10^{\{-3\}}$	1

Resistor Datasheet

Color	1st Band	2nd Band	3rd Band	Decimal Multiplier	Decimal Multiplier	Tolerance
Black	0	0	0	1	1	
Brown	1	1	1	10	10	±1 %
Red	2	2	2	100	100	±2 %
Orange	3	3	3	1K	1,000	
Yellow	4	4	4	10K	10,000	
Green	5	5	5	100K	100,000	
Blue	6	6	6	1M	1,000,000	
Violet	7	7	7	10M	10,000,000	
Gray	8	8	8		100,000,000	
White	9	9	9		1,000,000,000	
Gold					0.1	±5 %
Silver					0.01	±10 %

Battery Calculator Section’s Datasheet

Section / Tool	Output	Inputs	Formulas used
1. Conversions	Voltage (V)	Watt-hours & Amp-hours	$V = Wh / Ah$
	Watt-hours (Wh)	Amp-hours & Voltage	$Wh = Ah * V$
	Amp-hours (Ah)	Watt-hours & Voltage	$Ah = Wh / V$
2. Battery Life	Runtime (Hours)	Capacity & Current Load	$Hours = Capacity\ (mAh) / Load\ (mA)$
	Runtime (Minutes)	Capacity & Current Load	$Minutes = (Capacity\ (mAh) / Load\ (mA)) * 60$
	Runtime (Seconds)	Capacity & Current Load	$Seconds = (Capacity\ (mAh) / Load\ (mA)) * 3600$
3. Current Limit	Current (Amps)	Voltage & Resistance	$Current = Voltage / Resistance$
4. Voltage Sag	Voltage Drop (V)	Internal R & Current Load	$Voltage\ Drop = Internal\ Resistance\ (Ohms) * Load\ (A)$